

Ethics and Direction in Machine Learning Programs

Questions:

1. Is there personal information that may be exposed?
2. Are there regional or cultural biases in climate change that might be made worse by using machine learning?
3. Is there human bias in climate change that might be propagated while training machine learning?
4. Could machine learning potentially make incorrect decisions about where weather conditions might worsen and cause harm?

What are Ethics and Why Do They Matter? Machine Learning Edition

When considering the questions above, about how climate change data has the potential to have biases or ethical issues, there's a lot to review. Briefly studying the data from ClimateWins there are various parameters taken into consideration.

To answer the first question, I don't believe personal information may be exposed when looking at this data as it is presented. However, if there are data statistics taken from location availability of local residence in Europe in those countries, there is a potential their personal information is used and could be leaked.

Question two asks how machine learning can cause regional or cultural biases in climate change. When reading the Project Brief, it was explained that ClimateWins has been sorting through hurricane predictions from [The Japan Meteorological Agency \(JMA\)](#). If Machine Learning isn't properly monitored it could use the data from JMA to inappropriately predict terrible hurricane outcomes that could damage Japan's regional reputation or may be incorrect for Europe's region. In May 2024 Japan was ranked 3rd among travel destinations (read that article, [here](#)). If Machine Learning wrongly predicted weather patterns and the global broadcast of dangerous weather reports began circulating the news, Japan's tourist income could be drastically affecting, thus damaging that region.

I believe there is always potential for human bias whether we intentionally program for it or not. With the right weather reports and statistics, anyone can make data say climate change is happening quicker than it is. The analysts in question may believe they're helping the climate change situation by making it sound worse so more of the population starts to change their habits that are affecting the weather. However, this can also promote global panic if, for instance, you say temperatures are rising at a higher rate, people may be more inclined to believe glaciers will melt faster thus flooding the planet even quicker.

Machine Learning is just an algorithm and without monitoring it has the potential to make mistakes. It is created by humans who make mistakes. It's difficult to have 100% trust in machine learning as it's programmed to use skills to achieve our goals but it's not AI. Machine Learning only assists with what we design it to do it cannot make decisions to learn from its mistakes.